

Spring 2020

Problem 1 (Sp20). For which pairs of real numbers r, s does the following double integral converge?

$$\int_0^\infty \int_0^\infty \frac{dx dy}{1 + x^r y^s}$$

Problem 2 (Sp20, Fa21). Let $p(t) = t^n - p_1 t^{n-1} - p_2 t^{n-2} - \dots - p_n$ where all the coefficients p_j are positive. Show that p has exactly one positive zero.

Problem 3 (Fa77, Sp80, Sp20). Let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable for each $n = 1, 2, \dots$ with $|f'_n(x)| \leq 1$ for all n and x . Assume

$$\lim_{n \rightarrow \infty} f_n(x) = g(x)$$

for all x . Prove that $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous.

Problem 4 (Sp20). The Bernoulli numbers B_n are defined by

$$\sum_{n=0}^{\infty} \frac{B_n z^n}{n!} = \frac{z}{e^z - 1}$$

Find the largest integer N such that the sequence $\frac{B_n N^n}{n!}$ is bounded.

Problem 5 (Sp20). Is there an entire function taking the values $1, 1, 1/2, 1/2, 1/3, 1/3, \dots$ at the points $1, 1/2, 1/3, \dots$, respectively? Give an example or prove that no such function exists.

Problem 6 (Sp20). Let A be an $n \times n$ complex invertible matrix and define $X_{k+1} = 2X_k - X_k A X_k$ for integer $k \geq 0$. Show that X_k has a limit X as $k \rightarrow \infty$ if $\|I - AX_0\| < 1$ and identify the limit.

Problem 7 (Sp83, Fa21, Sp20, Fa20). Let A be an invertible real $n \times n$ matrix. Show that there is a decomposition $A = QR$ in which Q is an $n \times n$ real orthogonal matrix and R is upper triangular with positive diagonal entries. Is this decomposition unique?

Problem 8 (Sp20). Let $R = \mathbb{K}[x]/\langle x^n \rangle$ be a ring, where $\mathbb{K}$ is a field and n is a positive integer. Find all ideals of R .

Problem 9 (Sp20). Let $\mathbb{K}$ be a field. Denote by R the subring of the polynomial ring $\mathbb{K}[x]$ consisting of all $f(x) \in \mathbb{K}[x]$ such that $f(0) = f(1)$. Find a finite set of generators for R as an algebra over $\mathbb{K}$. This means that every element of R can be written as a polynomial in these elements with coefficients in $\mathbb{K}$.

Problem 10 (Sp20, Fa21). Find the unique continuous extension g of $f(x) = x^x$ to the half-open interval $\mathbb{R}_+ = [0, \infty)$ and determine all its fixed points there. Describe the limit x_∞ of the sequence $x_{n+1} = g(x_n)$ as a function of $x_0 \in \mathbb{R}_+$.

Problem 11 (Sp20). Let $J_n(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-int + ix \sin t} dt$ for $n \in \mathbb{Z}$. Show that

$$J_n(x + y) = \sum_{m \in \mathbb{Z}} J_{n-m}(x) J_m(y)$$

Problem 12 (Sp20). Suppose $n \geq 1$, and there are n real numbers a_j and n distinct real numbers t_j such that

$$\int_{-1}^1 p(t) dt = \sum_{j=1}^n a_j p(t_j)$$

for all polynomials p of degree at most $2n-1$. Show that $a_j > 0$ and $|t_j| \leq 1$ for $j = 1, \dots, n$.

Problem 13 (Fa85, Sp20). Evaluate $\int_0^{2\pi} e^{e^{i\theta}} d\theta$.

Problem 14 (Sp77, Sp82, Sp20). Prove the Fundamental Theorem of Algebra: Every nonconstant polynomial with complex coefficients has a complex zero.

Problem 15 (Sp20). Put $u = (1, 2, 3)$ and let

$$A = u^T u = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$$

Compute A^n for all integers $n \geq 1$ and find the largest eigenvalue of A .

Problem 16 (Sp20, Fa20, Fa25). Let V be a vector space of dimension n over a finite field with q elements. Prove that the number of subspaces $W \subseteq V$ of dimension k is equal to

$$\frac{\prod_{j=1}^n (q^j - 1)}{\left(\prod_{j=1}^k (q^j - 1)\right) \left(\prod_{j=1}^{n-k} (q^j - 1)\right)}$$

Problem 17 (Sp20). Let p be a prime number. View $\mathbb{Z}/p\mathbb{Z}$ as a subgroup of $(\mathbb{Z}/p^5\mathbb{Z}) \oplus (\mathbb{Z}/p^5\mathbb{Z})$ by the homomorphism sending 1 to (p^4, p^4) . Prove that the quotient of $(\mathbb{Z}/p^5\mathbb{Z}) \oplus (\mathbb{Z}/p^5\mathbb{Z})$ by $\mathbb{Z}/p\mathbb{Z}$ is isomorphic to $(\mathbb{Z}/p^a\mathbb{Z}) \oplus (\mathbb{Z}/p^b\mathbb{Z})$ for some integers a, b with $a \leq b$. Find a and b .

Problem 18 (Sp20). Let $f : \mathbb{Z}^n \rightarrow \mathbb{Z}^n$ be a homomorphism. Prove that f is an isomorphism if and only if the homomorphism $f_p : (\mathbb{Z}/p\mathbb{Z})^n \rightarrow (\mathbb{Z}/p\mathbb{Z})^n$ induced from f by reduction modulo p is an isomorphism for all prime numbers p .